

Lecture 9: Applied Regression in Economics

Credibility, Diagnostics, and Knowing When to Worry

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Where we are

What you can already do:

- Pull data via APIs (Census, FRED)
- Clean, merge, reshape
- Visualize and write reproducible notebooks

Today: use regression like an applied economist — not just run it, but *stress-test it*.

Next week: causal inference (when OLS breaks down by design).

Today's question

How do you know whether to believe a regression result?

Open question — vote before we start

The twins puzzle

A standard OLS regression of log wages on years of schooling returns roughly **8–10%** per year of education.

Studies of **identical twins** with different schooling levels find a return closer to **4–6%**.

Where did the other 4–5 percentage points go?

- Is one study wrong?
- Are identical twins special?
- Is something systematically inflating the OLS estimate?

We will answer this precisely by the end of class.

Learning objectives

By the end of today you will be able to:

- ① Interpret OLS coefficients *in context* — units, magnitudes, elasticities.
- ② Choose functional forms (level–level, log–level, log–log) *intentionally*.
- ③ Explain omitted variable bias (OVB) *with a number*, not just a story.
- ④ Recognize heteroskedasticity and know when robust SE changes your conclusions.
- ⑤ Read a **coefficient stability plot** — a standard applied robustness check.
- ⑥ Write a 3-sentence brief that a non-economist can act on.

About today's data

Calibrated synthetic data — not made up, not real.

The parameters we planted into the data-generating process come from published labor economics estimates:

- $\approx 7.5\%$ return to education (CPS estimates)
- $\approx 10\%$ union wage premium (Card, 1996)
- $\approx 7\%$ raw gender gap (IPUMS CPS)
- Diminishing returns to experience (Mincer, 1974)

Why synthetic?

In real data you never know the true coefficient.

Today we *do* know it. That's the whole point.

OLS — the part you already know

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

OLS chooses $\hat{\beta}$ to minimize $\sum_i \hat{u}_i^2$.

What OLS does well:

- Best linear unbiased estimator (BLUE) under classical assumptions
- Familiar, interpretable, fast
- Works as a building block for more complex models

What can go wrong:

- Omitted variables $\Rightarrow$ biased $\hat{\beta}$
- Wrong functional form $\Rightarrow$ wrong interpretation
- Heteroskedasticity $\Rightarrow$ wrong standard errors
- Influential observations $\Rightarrow$ unstable estimates

In applied work, the hard part is not running OLS. It is knowing which of these you have.

Interpretation: always start with units

Exercise — before you run anything

You are about to estimate:

$$\text{EmploymentRate}_i = \beta_0 + \beta_1 \text{MinimumWage}_i + u_i$$

What units is β_1 in? What would a “large” vs. “small” estimate look like?

Discuss with your neighbor for 90 seconds.

The answer depends entirely on:

- Is EmploymentRate in % points (0–100) or a share (0–1)? **Factor of 100.**
- Is MinimumWage in nominal or real dollars?
- Is the sample one state, all states, one quarter, or ten years?

Rule: write the units of every coefficient before interpreting the number.

Functional forms — choose, don't guess

Form	Model	β_1 means ...
Level–Level	$Y = \beta_0 + \beta_1 X$	ΔY per 1 unit ΔX
Log–Level	$\ln Y = \beta_0 + \beta_1 X$	$\approx 100\beta_1\%$ ΔY per 1 unit ΔX
Log–Log	$\ln Y = \beta_0 + \beta_1 \ln X$	$\% \Delta Y$ per 1% ΔX (elasticity)

How to choose:

- Skewed outcome? $\Rightarrow \log Y$
- Non-linear relationship? $\Rightarrow \log X$ or polynomial
- Want elasticity? $\Rightarrow \log\text{-}\log$

More precise log rule

$$\% \Delta Y = 100 \cdot (e^{\beta_1} - 1)$$

Matters when $|\beta_1| > 0.1$

Omitted Variable Bias — with a number

Suppose the true model is:

$$Wage_i = \beta_0 + \beta_1 Educ_i + \beta_2 Ability_i + u_i$$

You can observe *Educ*, but not *Ability*. You estimate:

$$Wage_i = \alpha_0 + \alpha_1 Educ_i + v_i$$

OVB formula:

$$\text{plim } \hat{\alpha}_1 = \beta_1 + \underbrace{\beta_2 \cdot \delta_1}_{\text{bias}}$$

where δ_1 = coefficient from regressing *Ability* on *Educ*.

	Sign of β_2	Sign of δ_1
Upward bias	+	+
Downward bias	+	—

Today's exercise: compare a case where robust SE changes a conclusion vs. where it does not. They are not always different — knowing when it matters is the skill.

OLS coefficients remain **unbiased** under heteroskedasticity.

But standard errors (and therefore t-stats, p-values, CIs) can be **wrong**.

Why it is common in economics:

- Wages and incomes (right-skewed)
- Firm size
- County or state populations
- Anything with a natural floor of zero

Applied default

Always report **robust standard errors** (HC1).

The coefficients do not change.

The inference might.

Robust SE in Python

```
import statsmodels.formula.api as smf

# Conventional SE
m = smf.ols("log_wage ~ educ + exper + union + female", data=df).fit()

# Robust SE (HC1 -- standard in labor economics)
m_rob = smf.ols("log_wage ~ educ + exper + union + female", data=df).fit(cov_type="HC1")

# Same coefficients, potentially different SE / p-values
print(m.params.compare(m_rob.params)) # coefficients: identical
print(m.bse.compare(m_rob.bse)) # standard errors: may differ
```

Rule of thumb: if the robust SE is noticeably larger, your original inference was overconfident.

Coefficient stability — the robustness check

One of the most useful informal diagnostics in applied work:

- 1 Estimate a **baseline** model (minimal controls).
- 2 Add controls progressively.
- 3 Plot the coefficient of interest with 95% CI at each step.

Stable coefficient:

Adding controls moves the estimate slightly but CIs overlap.

⇒ Result is not driven by omitted variables you can observe.

Unstable coefficient:

Adding one control changes the estimate dramatically.

⇒ Strong signal of OVB or multicollinearity. Worry.

We will build this plot in class today for the education coefficient.

Today's three-act workflow

Act	What you do	Goal
I	EDA + commit to a model	Form expectations; justify choices
II	Break the regression	Discover where it fails and why
III	Report honestly	Write for a practitioner audience

At each act, you will update a **living executive summary** — one sentence that evolves as you learn more. By the end you will have a defensible 3-sentence brief.

Act I — Explore and commit

- 1 Look at the wage distribution. Should you model $wage$ or $\ln(wage)$? Why?
- 2 Run a simple scatterplot of wages vs. education. Does the relationship look linear?
- 3 Before running any regression: write down what sign you expect for each coefficient and why.

Stop and write (2 minutes)

In your notebook, write your prior:

- Education coefficient: positive / negative, and roughly what size?
- Experience coefficient: do you expect diminishing returns?
- Union premium: what do you expect and why?

Act II — Break the regression

Exercise A: Subsample stability

Run your preferred model on:

- Workers with ≤ 12 years of education (HS or less)
- Workers with ≥ 16 years of education (college+)

Does the gender coefficient behave the same in both groups? If not, what explains it?

Exercise B: OVB vote

Before adding ability to the model, write your prediction:

- Will the education coefficient go up, down, or stay the same?
- By how much (as a fraction of its current value)?

Then run it. Were you right?

Diagnostics — a practical checklist

- ① **Residuals vs. fitted:** random scatter = OK; fan or curve = problem
- ② **Scale–location plot:** constant spread = homoskedastic
- ③ **Cook's distance:** influential points that shift your coefficients
- ④ **Coefficient stability:** your estimate across nested models

Goal: build the habit of checking before you report.

R^2 reminder

High R^2 does not mean causal.

Low R^2 does not mean useless.

R^2 measures *fit*, not *credibility*.

Act III — Report honestly

A 3-sentence executive brief for a non-economist:

- ① **Finding:** state the main relationship with magnitude and units.
- ② **Uncertainty:** quantify it; state whether the result is causal or associational.
- ③ **Limitation:** name the most important threat to validity.

Example

An additional year of schooling is associated with approximately a 7% increase in wages in this sample, holding experience and union status constant (robust 95% CI: 6.2–7.8%). This estimate should be interpreted as associational, not causal, as unobserved factors like ability are correlated with both education and wages. Were we able to observe ability directly, the estimate drops to around 5%, suggesting meaningful upward bias in the baseline result.

Answering the opening question

The twins puzzle — revisited

OLS of log wages on education: $\hat{\beta}_1 \approx 8\text{--}10\%$

Identical twins studies: $\approx 4\text{--}6\%$

We can now be precise:

$$\hat{\alpha}_1^{\text{OLS}} = \underbrace{\beta_1^{\text{causal}}}_{\approx 5\%} + \underbrace{\hat{\beta}_2^{\text{ability}} \cdot \hat{\delta}_1^{\text{ability on educ}}}_{\approx 3\text{--}5\%}$$

Identical twins share genetic endowment, so differencing within pairs removes most of the ability confound — that is what reduces the estimate.

This is not a curiosity. It is the central challenge of empirical economics.

Deliverable (end of class)

Your notebook should contain:

- **Prior** — written predictions before running each model.
- **At least 2 regressions** with interpretation (correct units, correct magnitude language).
- **Robust SE** for your preferred model; note whether it changes inference.
- **OVB demo** — comparison table with your commentary on the direction and size of bias.
- **Coefficient stability plot.**
- **One diagnostic plot** with written interpretation.
- **3-sentence executive summary** (final version).

Today: we showed OVB exists and can be quantified when we observe the confounder.

Reality: we almost never observe it.

Next week — Causal Inference:

- What makes a good research design?
- Difference-in-Differences intuition
- Why a natural experiment beats more controls

Homework

Replicate today's log-wage regression on real CPS microdata (IPUMS extract).

Do the planted coefficients hold?

Post your stability plot to GitHub.